

ITAI-2376

MD_REPORT

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. Implementation Summary

This project uses the MNIST dataset of handwritten digits to construct a lightweight diffusion model.

The model performs the forward (noise adding) and reverse (denoising) operations common to diffusion models using a condensed U-Net architecture.

To illustrate fundamental generative notions, a lightweight training setup was employed.

The notebook also includes a **CLIP evaluation** stage to assess the semantic similarity between generated images and their corresponding text prompts (e.g., “digit 0”, “digit 1”, etc.).

Key components implemented:

- Dataset preprocessing and normalization
- Custom U-Net with GELUConvBlock, skip connections
- Linear noise schedule and forward diffusion
- Reverse diffusion for image sampling
- Visualization of generated samples
- CLIP-based evaluation for semantic quality

2. Generated Image Results

Generated MNIST digits using the trained diffusion model:

Observations:

- Digits are recognizable and show correct structural patterns.
- Noise reduction is progressive across timesteps.
- Variability exists in fine details due to lightweight training.

3. CLIP Evaluation Results

Figure 2. CLIP similarity scores for generated MNIST digits:

Observations:

- CLIP scores are higher for easily distinguishable digits (e.g., 0–3).
- Lower similarity for complex digits like 5 or 8 suggests model uncertainty.
- Confirms diffusion model learns semantically coherent representations.

4. Assessment Questions

1. Using your own words and illustrations, describe the forward diffusion process. At each timestep, the forward process progressively introduces Gaussian noise into the image, converting a clear image into pure noise. During training, this instructs the model to reverse the noise.
2. Talk about the effects of noise on learning and why it is introduced gradually. Instead of learning a direct mapping from noise to image, the model can learn local, incremental denoising by gradual noise addition, which increases stability.

Examine when the images become identifiable after the denoising process. Digit outlines start to show up clearly after 20 to 30 percent of the reverse steps.

Describe why diffusion models are a good fit for the U-Net architecture. When learning hierarchical representations, U-Net maintains spatial detail through skip connections, which is essential for reconstructing fine picture characteristics.

Explain the function and significance of skip connections. They avoid information loss during downsampling by moving low-level features from encoder to decoder layers.

Describe in detail how the class conditioning mechanism operates. Class embeddings could be concatenated with timestep embeddings to bias production toward particular numbers or introduced as additional input channels for future enhancement.

C. Training Analysis

1. Examine the information provided by your model's loss value. The accuracy of predicted noise compared to actual noise is measured by loss. More accurate denoising capability is shown by lower loss.
2. Explain how the quality of the generated images changed during training. Outputs were initially random noise. Basic digit contours were visible after a few batches.
3. Describe the function of diffusion models' temporal embedding. Time embeddings enable the model to apply the appropriate noise-removal intensity by indicating which stage of the diffusion process it is denoising.

D. CLIP Evaluation

1. Examine the information provided by CLIP scores regarding your created photos. They gauge how well the semantic content of an image corresponds to its intended label. Images with higher scores are more realistic and consistent with their class.

2. Explain why some images are simpler or more difficult to create. While more complicated numbers, like 5 or 8, have lower similarity scores, simpler ones, like 0 or 1, have consistent shapes.
3. Explain how CLIP scores could enhance the generation process. To promote semantic alignment, CLIP feedback might be employed as an auxiliary loss function.

E. Practical Applications

5Talk about possible practical uses.

Digit synthesis for generative art, anomaly detection, image restoration, and data augmentation.

Examine the current model's shortcomings.

Realism is constrained by a tiny dataset, limited resolution, and short training time.

Name three enhancements and explain them.

For better temporal awareness, include timestep embeddings.

For higher-quality reconstruction, use a deeper U-Net.

Use classifier-free guidance to ensure that sampling is regulated.

. Lessons Learned

- Diffusion models offer controllable, stable generative capabilities.
- Even lightweight models can learn meaningful visual priors.
- CLIP evaluation provides a powerful semantic quality check.